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Final Report

Vehicle Classification Using Conventional and CNN Methods

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Submitted By

Name: Faria Ahmed

ID: 2125051087

Department: CSE

Section: 7B2

Batch: 50

Submitted To

Mrinmoy Biswas Akash

Lecturer & Course Coordinator

M.Sc. (Engg.) in CSE, Jatiya Kabi Kazi Nazrul Islam University (JKKNIU)

Ratri Datta

Lecturer

B.Sc. in CSE, Pabna University of Science and Technology (PUST)

Vehicle Classification Using Conventional and CNN Methods

Introduction

This work aims to classify images of vehicles as cars, trucks, buses, and motorcycles. In this work, the approach towards the task was performed in two ways: by traditional machine learning (Support Vector Machine, SVM) and by deep learning (Convolutional Neural Network, CNN). The project compares both techniques for their strengths and weaknesses to handle image classification.

Two approaches were explored:

Conventional Machine Learning (SVM): A baseline to understand traditional methods of image classification.

Deep Learning (CNN): A modern approach utilizing Convolutional Neural Networks to leverage automated feature extraction for improved accuracy.

Dataset Description

Source: The dataset was obtained from https://www.kaggle.com/datasets/kaggleashwin/vehicle-type-recognition.

2.1 Source

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The dataset used is the publicly available from Kaggle. It consists of images labeled into four classes:

- **Car**
- **Truck**
- **Bus**
- **Motorcycle**

2.2 Dataset Structure

- **Number of Classes:** 4
- **Number of Images:** Approximately 3,000.
- **Format:** Images are stored in subfolders corresponding to their class.

2.3 Preprocessing

To prepare the data for analysis and modeling:

1. **Resizing:** All images were resized to a uniform size of 128x128 pixels to reduce computational requirements.
2. **Normalization:** Pixel values were scaled to the range [0, 1].
3. **Encoding:** Labels were converted to numeric integers for compatibility with machine learning and deep learning models

Methodology

3.1 Support Vector Machine (SVM)

The SVM model was implemented as a conventional machine learning approach:

1. **Feature Extraction:**
    - Images were flattened into 1D arrays, losing spatial information but retaining pixel intensity.
2. **Model:**
    - A linear kernel SVM was trained using the extracted features.
3. **Training and Testing:**
    - The dataset was split into training and testing subsets (80/20 split).
4. **Evaluation:**
    - Classification accuracy and a confusion matrix were used to measure performance.

3.2 Convolutional Neural Network (CNN)

CNN was implemented as a deep learning approach:

1. **Architecture:**
    - **Convolutional Layers:** Extract spatial features using filters.
    - **Max-Pooling Layers:** Reduce spatial dimensions to retain essential features.
    - **Dense Layers:** Perform classification based on extracted features.
    - **Softmax Activation:** Used for multi-class classification.
2. **Regularization:**
    - Dropout layers were added to prevent overfitting.
3. **Training:**
    - Data augmentation (rotation, zoom, flipping) was applied to improve generalization.
    - The model was trained for 10 epochs using the Adam optimizer and categorical crossentropy loss.
4. **Evaluation:**
    - Accuracy, loss, and confusion matrix were used to assess performance.

Results and Discussion

GitHub Link: https://github.com/faria087/ML-Lab-project

4.1 Training Performance

SVM:

- **Training Accuracy:** ~70%
- **Test Accuracy:** ~50%
- **Observations:**
    - Flattening images into 1D arrays caused loss of spatial information, limiting SVM's ability to learn meaningful patterns.
    - Performance was low, highlighting the limitations of conventional methods for image classification.

CNN:

- **Training Accuracy:** 96.7%
- **Validation Accuracy:** 65.6%
- **Test Accuracy:** ~65%
- **Observations:**
    - CNN achieved much higher accuracy due to its ability to learn spatial and hierarchical features from image data.
    - Regularization via dropout and data augmentation reduced overfitting.

4.2 Confusion Matrix

The confusion matrix for the CNN model revealed the following:

- **Strengths:**
    - High accuracy in recognizing classes like "Car" and "Motorcycle."
- **Weaknesses:**
    - Misclassifications occurred between "Bus" and "Truck," likely due to similar visual features.

4.3 Comparison Analysis:

- CNN outperformed SVM by a significant margin due to its ability to extract spatial features from image data.

Metric	**SVM**	**CNN**
Test Accuracy	~50%	~65%
Training Time	Fast	Moderate
Feature Extraction	Manual	Automated
Generalization	Low	High

Conclusion

This project highlighted how deep learning methods, namely CNN, outperform some of the traditional methods-SVM-on image classification tasks; whereas the SVM provided merely a baseline, the accuracy was much higher for CNN because it leveraged spatial feature extraction.

Key Findings:

- CNN achieved a test accuracy of ~65%, demonstrating its capability to handle image classification tasks effectively.
- Data augmentation improved generalization by increasing the diversity of training samples.
- SVM, while simple, was not suitable for high-dimensional image data due to its inability to capture spatial features.

Future Work:

- Increase dataset size or do more augmentation to improve generalization.
- Experiment with deeper CNN architectures and hyperparameter tuning.
- Deploy the model in real-world applications, for example, traffic monitoring or autonomous vehicle systems.